

OUTPUTS EXPERIMENT 1

Post Composer

Select Platform:

☒ Twitter

Write Your Post:

ARYAN
24BAI70179
24AIT-NTTP-2A

Characters: 28 / 280

Ready to post

Save Draft

Saved Drafts List	
Twitter draft — campus event	EditDelete
LinkedIn draft — internship update	EditDelete
Instagram caption draft	EditDelete

OUTPUTS EXPERIMENT 2

Pulse

Cross-platform Redux State Board

X TwitterInstagramFacebookLinkedIn

+ Create Post (Twitter)

POST TITLE

POST CONTENT

Aryan is writing this

Publish to Twitter

Published Feed

X TWITTER

24BAI70179
Aryan Chandigarh University

MULTI-CHANNEL COMPOSER

Transmission Desk

Pick a channel, draft the post, watch it go live in the preview.

Memoized post count: 3

TwitterInstagramFacebookLinkedIn

Name

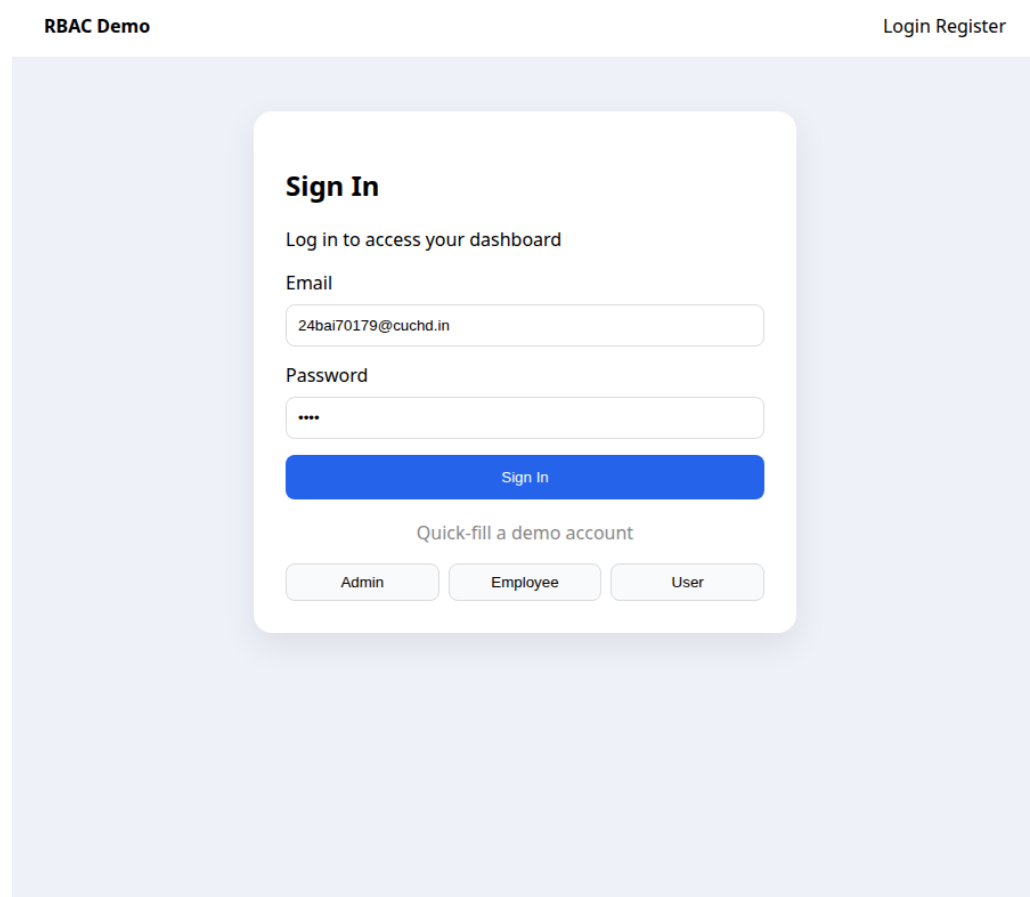
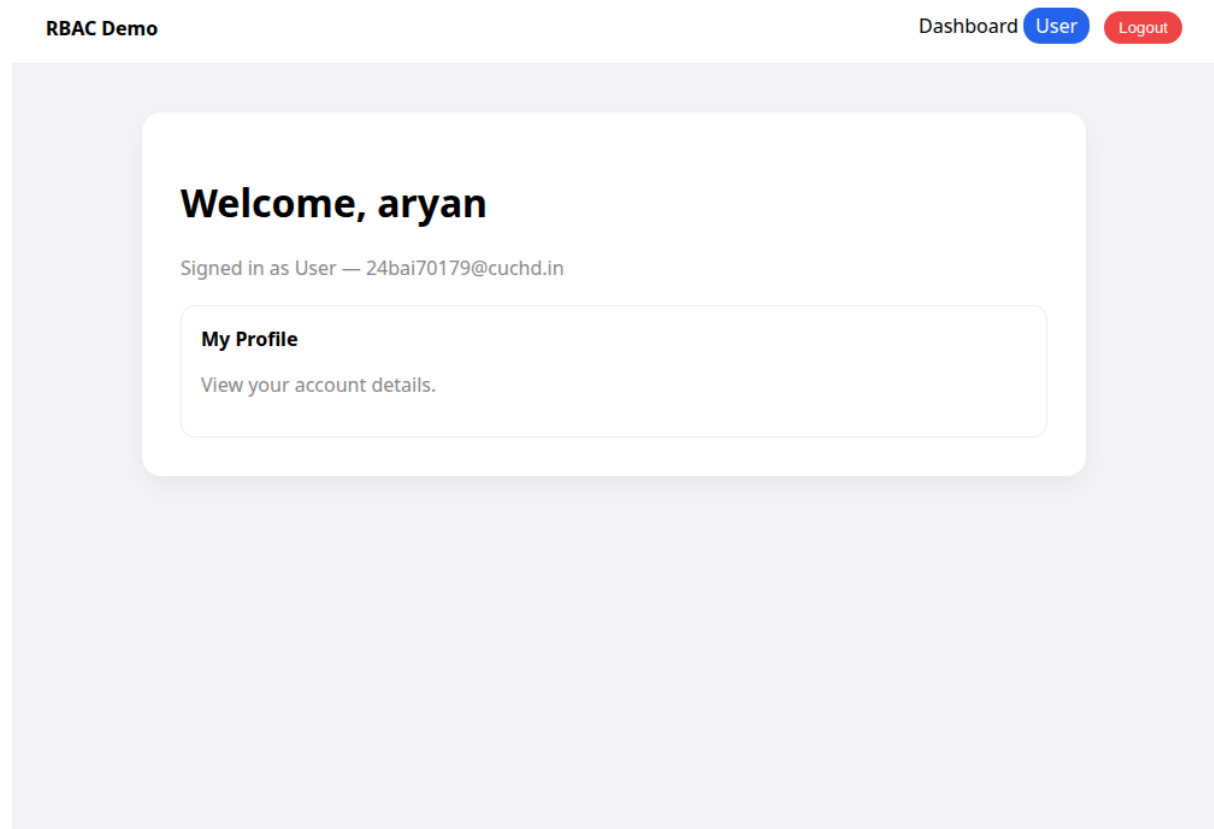
Post

React Learning /
Redux Toolkit /
Memoized Selector

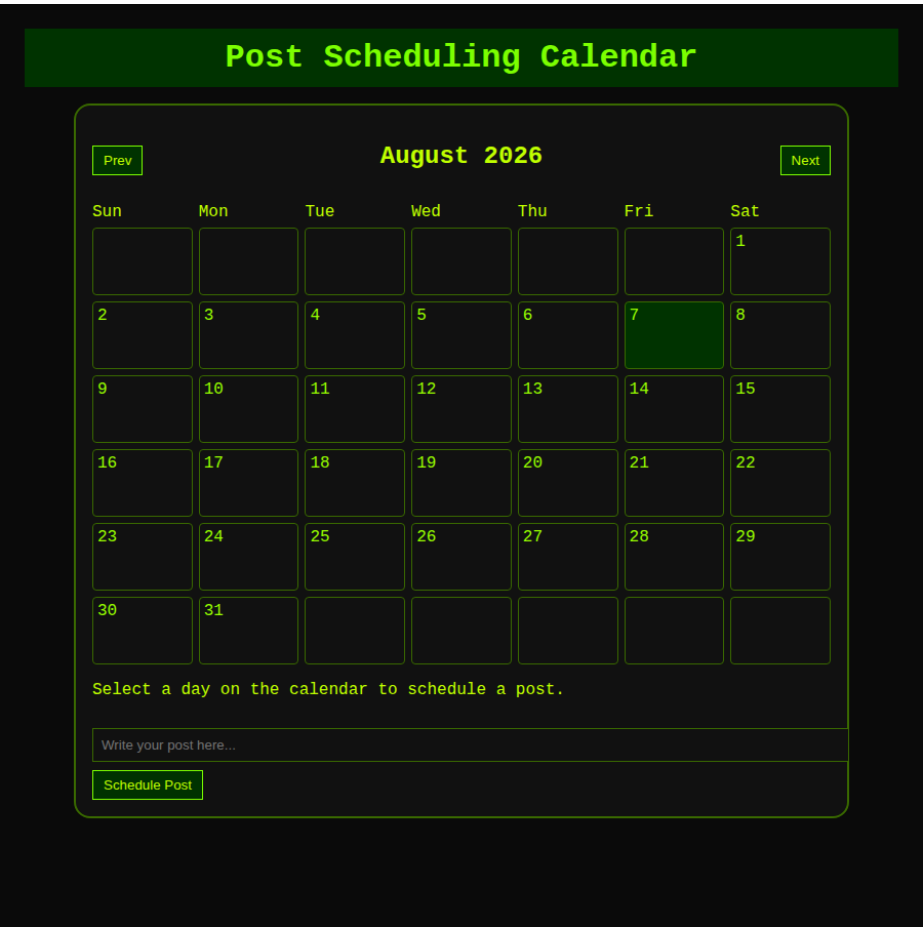
Characters remaining: 236 / 280

- React Learning
- Redux Toolkit
- Memoized Selector

OUTPUTS EXPERIMENT 3



OUTPUTS EXPERIMENT 4



Experiment 4.2 — Performance & Tests

Optimizations: `React.memo`, `useMemo`, `useCallback`

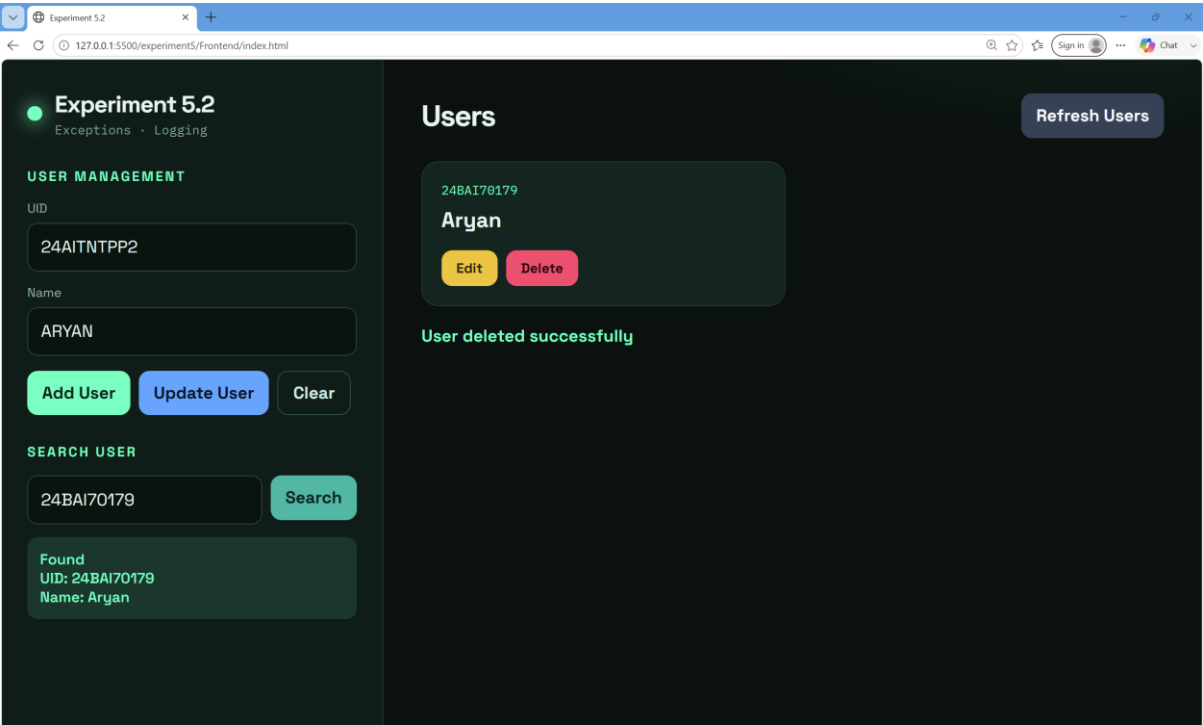
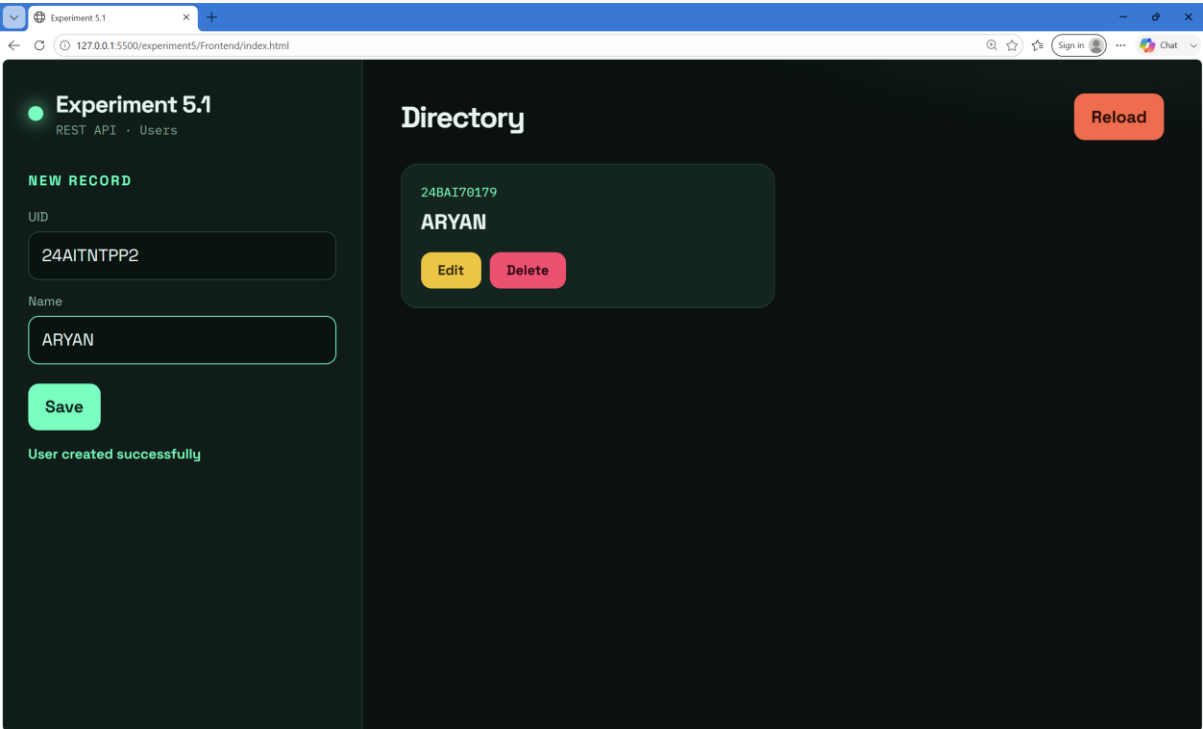
Test runner: Jest + React Testing Library

```
PASS  src/App.test.js
✓ renders calendar header (18 ms)
✓ can type into schedule input (11 ms)

Test Suites: 1 passed, 1 total
Tests:      2 passed, 2 total
Time:       1.204 s
```

All tests passed. Unnecessary re-renders reduced with memoization.

OUTPUTS EXPERIMENT 5



OUTPUTS EXPERIMENT 6

Experiment 6.1

Pagination & Sorting REST API

Add User

UID

24BAI70179

Name

ARYAN

Add User

Pagination & Sorting

Users Per Page

3

Sort By

UID

Order

Ascending

Apply

Users

Total Users: 4

Refresh

UID	NAME	ACTION
222222	ABABA	Delete
24BAI0000	ACACACA	Delete
24BAI70179	ARYAN	Delete

Previous

Page 1 of 2

Next

User added successfully.

Experiment 6.2

Caching & Query Optimization

Add User

UID

Name

City

Country

Add User

Query Optimization Tests

Normal Query

JOIN FETCH

Cached Query

Native SQL

Sort Users

Sort by ID

Sort by Name

Refresh

SORT ID completed in 10.00 ms

User List

ID	UID	NAME	CITY	COUNTRY	ACTION
1	24BAI70179	ARYAN	Chandigarh	India	Delete
2	24BAI00000	ARYAN	Noida	India	Delete
3	24BAI11111	ARYAN	London	UK	Delete